

Wing Area to Wing Length Ratio

Semion D

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1 Introduction

Let's say we have 2 paper airplanes. One of them is really long but narrow, while the other is short but wide. Both of them have the same wing area. The question is: which one will fly longer and over a greater distance, and which one will be more effective? This is the problem I will investigate in this short essay. It will be divided into 3 main parts: the Hypothesis, which will make a prediction of the outcome based on math, physics, and logic; next is the Experiment, which will test the hypothesis; and the last one is the Conclusion, which will summarize everything in a simple logic-math statement.

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2 Hypothesis

2.1 Forces

First, we need to define the forces that act upon this complex airplane-like object.

Here is the list:

Force	Representation
Gravity	Representation through constant $g = 9.81$ and mass of the object
Lift	Representation in Newtons as a variable L in formulas
Drag	Representation in Newtons as a variable D in formulas
Thrust	Representation in Newtons as a variable T in formulas

In this case, we will not use thrust for several reasons: it does not affect the efficiency of the airplane itself, it just creates a continuous kinetic energy input; and it is harder to simulate on a small experimental scale. The only situation in which thrust will be included is the power input when we throw the airplane. So, what's left is Gravity (or weight), Lift, and Drag.

Lift is a force that pushes the plane upward. It happens because the area under the wing has higher air pressure than the top, and this pressure difference moves the airplane up. It is calculated through this formula:

$$L = \frac{1}{2}\rho v^2 S C_L$$

Variable	Meaning	Unit
L	Lift	Newtons (N)
ρ	Air density	kg/m ³
v	Air velocity	m/s
S	Wing area	m ²
C_L	Coefficient of lift	Scalar

C_L is a scalar from 0 to 1 and combines in itself many wing parameters. One of the main ones is the angle of attack of the wing α , and it represents the angle of the wing. Here is the formula for it:

$$C_L = C_{L0} + a\alpha$$

a is the lift-curve slope, and in general, it tells us the speed at which C_L grows when α grows. Basically, it is the derivative of a linear function C_L . C_{L0} is the lift coefficient at 0° angle of attack, which might be positive or negative, but in this case, we will consider this equal to 0.

To calculate the lift-curve slope, we can use this formula:

$$a = \frac{a_0}{1 + \frac{a_0}{\pi e AR}}$$

where:

a = lift-curve slope

a_0 = 2D airfoil lift-curve slope

e = Oswald efficiency factor

AR = aspect ratio

a_0 tells you how quickly the lift coefficient C_L changes with angle of attack for an idealized wing cross-section extending infinitely in the spanwise direction (I was not expecting that I would write something so clever). But considering that the wing is a finite region, air starts to behave differently, creating friction and drag, which is included and considered in a . For our project, a_0 will be:

$$a_0 \approx 2\pi \text{ rad}^{-1}$$

Next is e , which tells us how efficiently a real finite wing produces lift compared with an ideal wing. It is a scalar from 0 to 1 and usually lies between 0.7 and 1. For this experiment, we will choose 0.7, which is usual for flat wings.

Our plane will also experience drag, which we cannot ignore, so here is the formula:

$$C_D = C_{D_p} + C_{D_i}$$

$$C_D = C_{D_p} + \frac{C_L^2}{\pi e AR}$$

And consequently:

$$D = \frac{1}{2} \rho v^2 S \left(C_{D_p} + \frac{C_L^2}{\pi e AR} \right)$$

Variable	Meaning	Unit
D	Total drag force	N
C_D	Total drag coefficient	Dimensionless
ρ	Air density	kg/m ³
v	Air velocity	m/s
S	Wing area	m ²
C_{D_p}	Parasite/profile drag coefficient	Dimensionless
C_L	Lift coefficient	Dimensionless
e	Oswald efficiency factor	Dimensionless
AR	Wing aspect ratio	Dimensionless

Before actually calculating the parameters of our 2 prototypes, we need to establish the difference between them using math. Engineers did this a long time ago and came up with this formula:

$$AR = \frac{b^2}{S}$$

AR is the aspect ratio, which represents the relationship between wing area S and wingspan b , which is the distance from one tip of the wing to another. This formula will help us to separate the 2 prototypes on a mathematical level.

2.2 Calculations for a Wide Airplane

We need to remember that we have 4 main forces acting upon this object.

$$\vec{L} = \mathbf{Lift}$$

$$\vec{D} = \mathbf{Drag}$$

$$\vec{T} = \mathbf{Thrust}$$

$$\vec{W} = \mathbf{Gravity}$$

We know that $\vec{L}$ is perpendicular to $\vec{D}$ and $\vec{T}$, but parallel to $\vec{W}$, so:

$$\vec{L} \perp \vec{D}$$

$$\vec{L} \perp \vec{T}$$

$$\vec{L} \parallel \vec{W}$$

So, if we think in terms of vector mathematics, we can write our net force of the flying airplane like this:

$$\vec{T} + \vec{D} + \vec{L} + \vec{W} = \vec{F}_{\text{net}}$$

$$\|\vec{F}_{\text{net}}\| = F_{\text{net}}$$

But we are not masochists to write it in vectors if we can write it just using scalars (which is less intuitive, but easier to calculate later). By knowing the vector orientations, we can form triangles and use the Pythagorean theorem:

$$F_{\text{net}} = \sqrt{(T - D)^2 + (L - W)^2}$$

Now we already have a basic pattern. For now, we need to substitute everything we know:

Therefore, the fully substituted equation is:

$$F_{\text{net}} = \sqrt{\left[T - \frac{1}{2}\rho v^2 S \left(C_{D_p} + \frac{S}{\pi e b^2} \left(\frac{a_0 \alpha}{1 + \frac{a_0 S}{\pi e b^2}} \right)^2 \right) \right]^2 + \left[\frac{1}{2}\rho v^2 S \left(\frac{a_0 \alpha}{1 + \frac{a_0 S}{\pi e b^2}} \right) - mg \right]^2}$$

Also, we can change T to the equation for force:

$$T = m \frac{\Delta v}{\Delta t}$$

$$F_{\text{net}} = \sqrt{\left[m \frac{\Delta v}{\Delta t} - \frac{1}{2}\rho v^2 S \left(C_{D_p} + \frac{S}{\pi e b^2} \left(\frac{2\pi \alpha}{1 + \frac{2S}{eb^2}} \right)^2 \right) \right]^2 + \left[\frac{1}{2}\rho v^2 S \left(\frac{2\pi \alpha}{1 + \frac{2S}{eb^2}} \right) - mg \right]^2}$$

Then we substitute:

$$C_{D_p} = 0$$

$$e = 0.7$$

$$a_0 = 2\pi \text{ rad}^{-1}$$

$$C_{L0} = 0$$

and standard near-sea-level values:

$$\rho = 1.225 \text{ kg/m}^3$$

$$g = 9.81 \text{ m/s}^2$$

Then:

$$\frac{1}{2}\rho = \frac{1}{2}(1.225) = 0.6125$$

and

$$\frac{a_0 S}{\pi e b^2} = \frac{2\pi S}{\pi(0.7)b^2} = \frac{2S}{0.7b^2} = \frac{20S}{7b^2}$$

the equation becomes:

$$F_{\text{net}} = \sqrt{\left[m \frac{\Delta v}{\Delta t} - 0.6125 v^2 S \left(\frac{S}{0.7\pi b^2} \left(\frac{2\pi\alpha}{1 + \frac{20S}{7b^2}} \right)^2 \right) \right]^2 + \left[0.6125 v^2 S \left(\frac{2\pi\alpha}{1 + \frac{20S}{7b^2}} \right) - 9.81mg \right]^2}$$

Then we need to choose the wingspan and area. For the wingspan, I will choose dimensions that are wide enough and easy enough to make on a 3D printer:

$$b = 0.30 \text{ m}$$

$$S = 0.0225 \text{ m}^2$$

So:

$$F_{\text{net}} = \sqrt{\left[m \frac{\Delta v}{\Delta t} - 0.6125 v^2 (0.0225) \left(\frac{0.0225}{0.7\pi(0.09)} \left(\frac{2\pi\alpha}{1.714286} \right)^2 \right) \right]^2 + \left[0.6125 v^2 (0.0225) \left(\frac{2\pi\alpha}{1.714286} \right) - 9.81mg \right]^2}$$

Then I designed an airplane (which you can find in this repo) and found that its weight is approximately 0.0205 kg. Which we can insert into the equation:

$$\sqrt{\left[0.0205 \frac{\Delta v}{\Delta t} - 0.6125 v^2 (0.0225) \left(\frac{0.0225}{0.7\pi(0.09)} \left(\frac{2\pi\alpha}{1.714286} \right)^2 \right) \right]^2 + \left[0.6125 v^2 (0.0225) \left(\frac{2\pi\alpha}{1.714286} \right) - 0.201 \right]^2}$$

And we can simplify it further.

Since:

$$0.6125(0.0225) = 0.01378125$$

we get:

$$\sqrt{\left[0.0205\frac{\Delta v}{\Delta t} - 0.01378125v^2 \left(\frac{0.0225}{0.063\pi} \left(\frac{2\pi\alpha}{1.714286}\right)^2\right)\right]^2 + \left[0.01378125v^2 \left(\frac{2\pi\alpha}{1.714286}\right) - 0.201105\right]^2} = F_D$$

Now we need to choose v , α and $\frac{\Delta v}{\Delta t}$.

$$v = 10 \text{ m/s}$$

$$\frac{\Delta v}{\Delta t} = 50 \text{ m/s}^2$$

Now we need to choose an angle of attack. Since we are making a plastic version of a paper airplane with flat wings, we can use a typical angle of attack of 5° for the model.

Therefore:

$$\alpha = 5^\circ = \frac{5\pi}{180} \approx 0.0873 \text{ rad}$$

First, we calculate the lift:

$$L = 0.01378125(10)^2 \left(\frac{2\pi(0.0873)}{1.714286}\right)$$

$$L \approx 0.44096 \text{ N}$$

Next, we calculate the induced drag:

$$D = 0.01378125(10)^2 \left(\frac{0.0225}{0.7\pi(0.09)} \left(\frac{2\pi(0.0873)}{1.714286}\right)^2\right)$$

$$D \approx 0.01604 \text{ N}$$

The thrust is:

$$T = (0.0205)(50) = 1.025 \text{ N}$$

and the weight is:

$$W = (0.0205)(9.81) = 0.201105 \text{ N}$$

Therefore, the horizontal and vertical components of the net force are:

$$F_x = T - D = 1.025 - 0.01604 = 1.00896 \text{ N}$$

$$F_y = L - W = 0.44096 - 0.201105 = 0.23986 \text{ N}$$

Finally:

$$F_{\text{net}} = \sqrt{(1.00896)^2 + (0.23986)^2}$$

$$F_{\text{net}} \approx 1.037 \text{ N}$$

In the next section, we will calculate F_{net} for a narrower airplane and compare them.

2.3 Calculations for a Narrower Airplane

The only thing that we will change here is b , which in turn will affect AR . S should stay the same. For b , I chose 20 cm. The mass is the same, as I was able to control it with the infill density of a 3D printer. Now what is left to calculate is F_{net} for a narrower airplane:

$$b = 0.20 \text{ m}$$

Therefore:

$$AR = \frac{b^2}{S} = \frac{0.20^2}{0.0225} \approx 1.7778$$

The denominator in the lift coefficient becomes:

$$1 + \frac{2S}{eb^2} = 1 + \frac{2(0.0225)}{0.7(0.20)^2} \approx 2.607143$$

Therefore, the net force equation becomes:

$$F_{\text{net}} = \sqrt{\left[0.0205 \frac{\Delta v}{\Delta t} - 0.6125 v^2 (0.0225) \left(\frac{0.0225}{0.7\pi(0.04)} \left(\frac{2\pi\alpha}{2.607143} \right)^2 \right)\right]^2 + \left[0.6125 v^2 (0.0225) \left(\frac{2\pi\alpha}{2.607143} \right)\right]^2}$$

Using:

$$v = 10 \text{ m/s}$$

$$\frac{\Delta v}{\Delta t} = 50 \text{ m/s}^2$$

and:

$$\alpha = 5^\circ = \frac{5\pi}{180} \approx 0.0873 \text{ rad}$$

we get:

$$L = 0.01378125(10)^2 \left(\frac{2\pi(0.0873)}{2.607143} \right)$$

$$L \approx 0.2898 \text{ N}$$

Next, we calculate the induced drag:

$$D = 0.01378125(10)^2 \left(\frac{0.0225}{0.7\pi(0.04)} \left(\frac{2\pi(0.0873)}{2.607143} \right)^2 \right)$$

$$D \approx 0.01559 \text{ N}$$

The thrust remains the same:

$$T = (0.0205)(50) = 1.025 \text{ N}$$

and the weight remains:

$$W = (0.0205)(9.81) = 0.201105 \text{ N}$$

Therefore:

$$F_x = T - D = 1.025 - 0.01559 = 1.00941 \text{ N}$$

$$F_y = L - W = 0.2898 - 0.201105 = 0.08873 \text{ N}$$

Finally:

$$F_{\text{net}} = \sqrt{(1.00941)^2 + (0.08873)^2}$$

$$F_{\text{net}} \approx 1.0133 \text{ N}$$

2.4 Hypothesis Conclusion

We calculated F_{net} for 2 planes with the same mass and wing area, but with a different b , which is the wingspan.

$$F_{\text{net,wide}} \approx 1.037 \text{ N}$$

$$F_{\text{net,narrow}} \approx 1.0133 \text{ N}$$

And we can conclude that:

$$F_{\text{net,wide}} \approx 1.037 > F_{\text{net,narrow}} \approx 1.0133$$

This means that the wide plane has a greater F_{net} right after launch, right after I release my plane, which will allow it to fly longer.

3 Experiment

3.1 Plane design

I tried many things before getting these planes to fly. The biggest problem was the center of mass. During the early stages of development, the planes twisted backward after launch and fell, like this one:

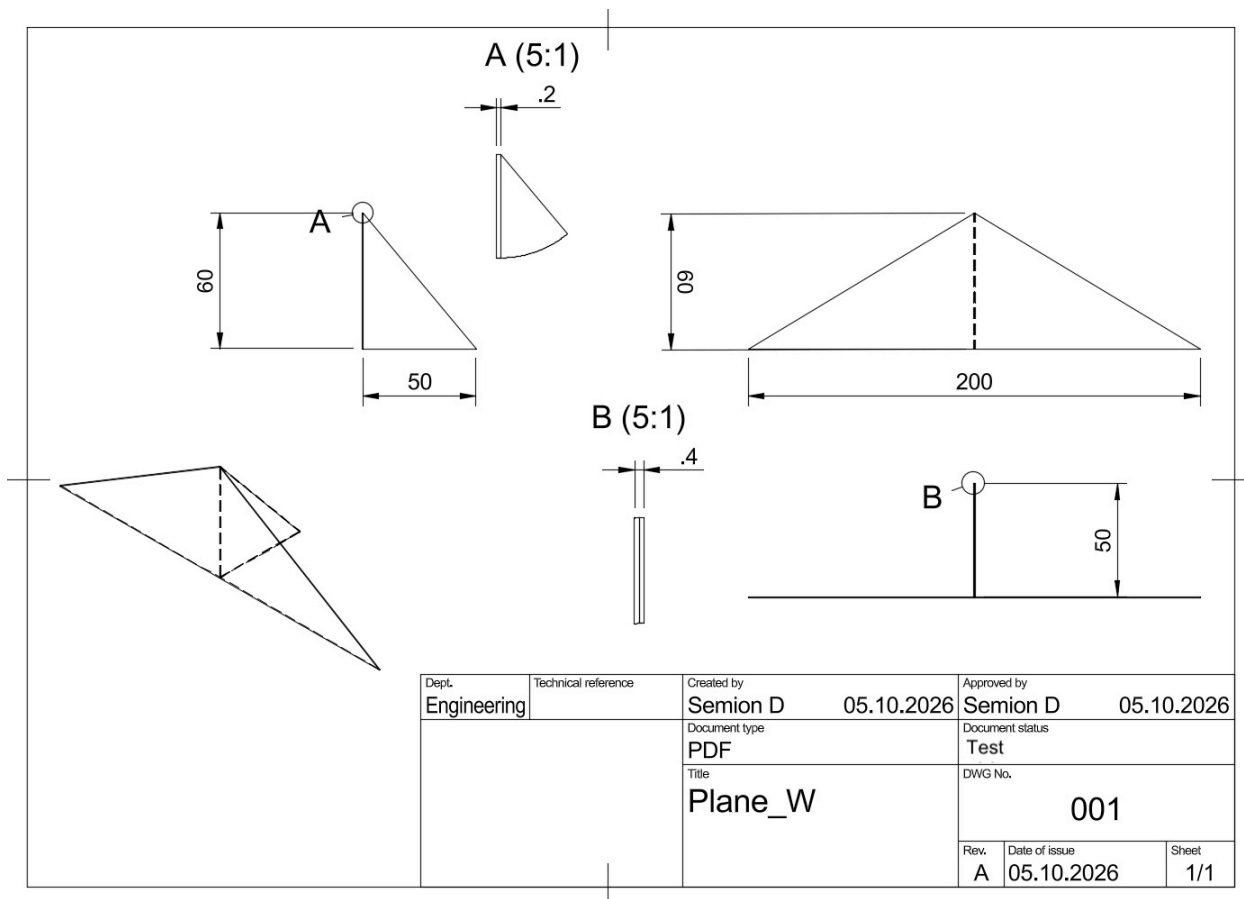


Figure 1: Plane W PDF – 001 Rev A.

The center of mass here is located behind the center of a $y - z$ plane slice, and rushing air forces the back to fall and the front to rise, twisting the plane. And that's why I added more material to the front, in order to move the center of mass to the front and stabilize the plane during the flight. And this is the result of my fight with engineering and nature:

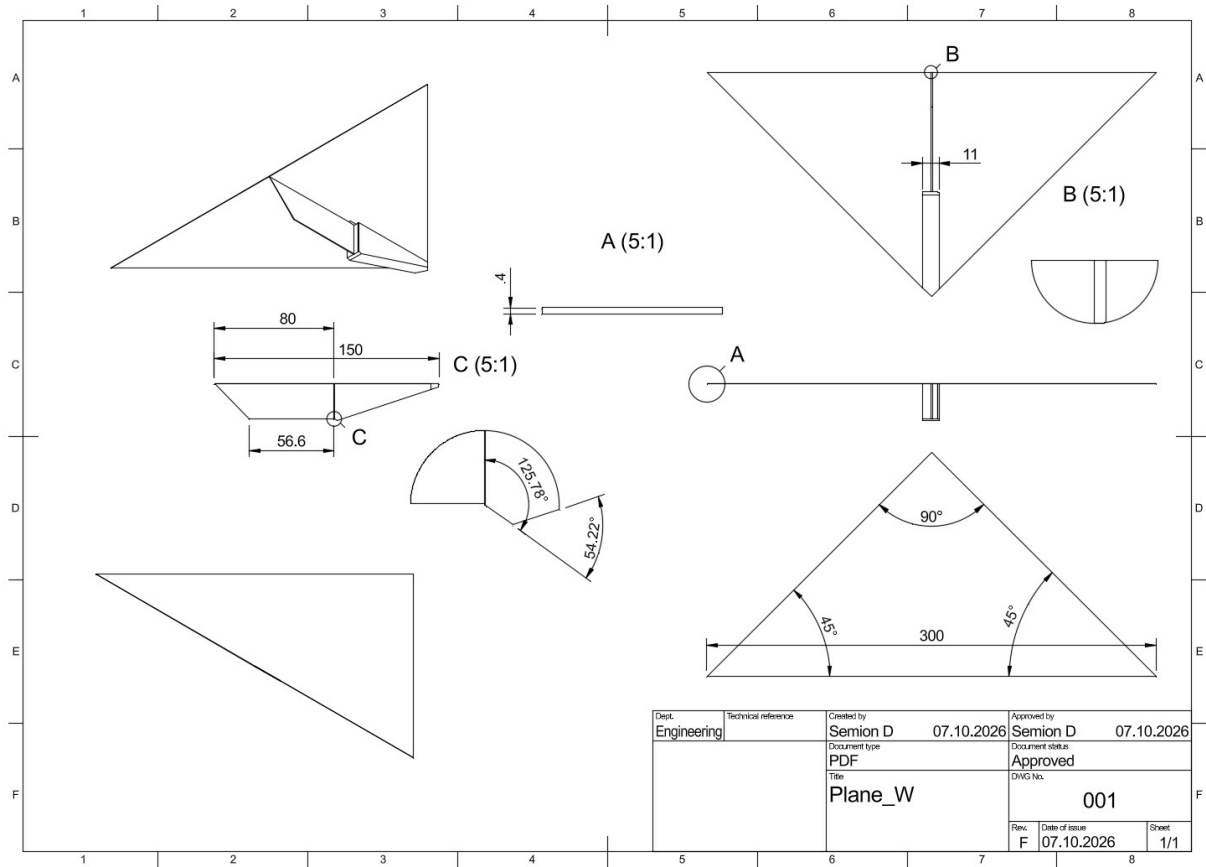


Figure 2: Plane W PDF – 001 Rev F (Wide Plane).

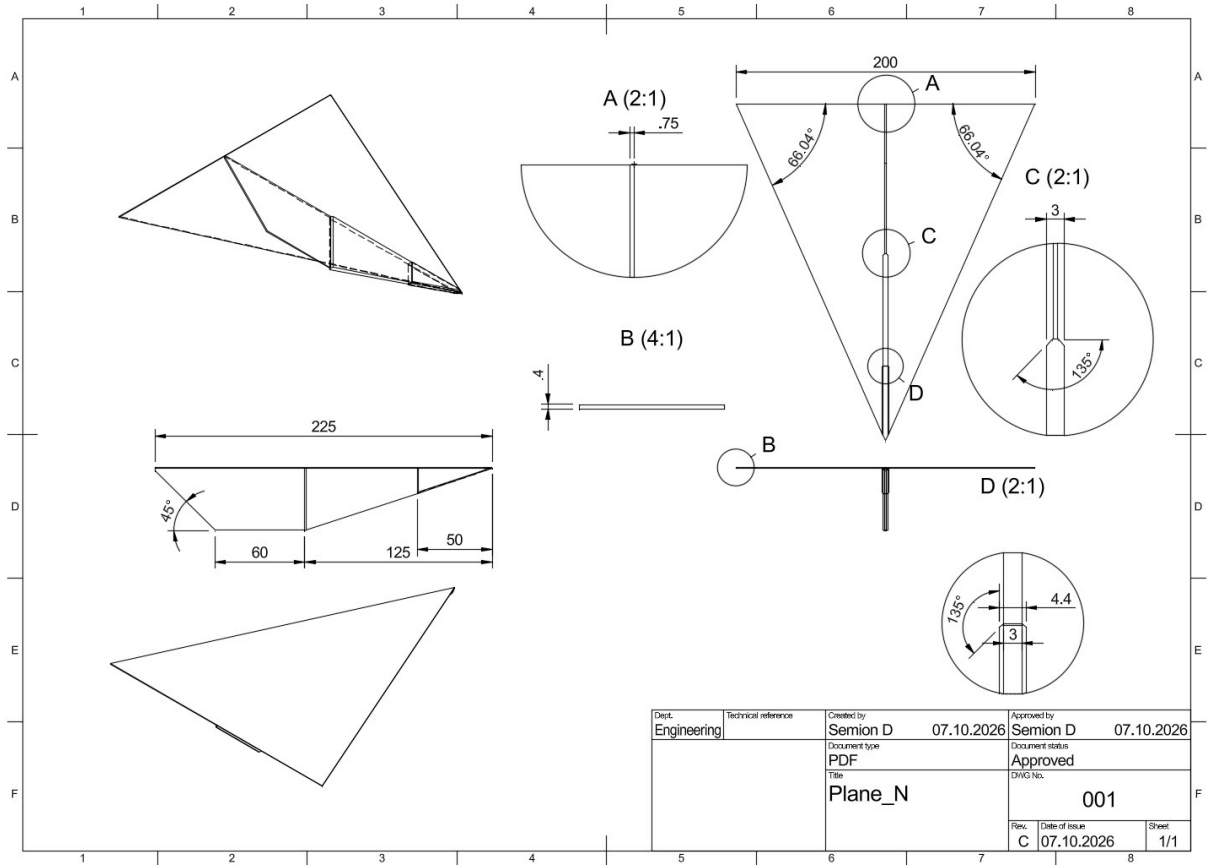


Figure 3: Plane N PDF – 001 Rev C (Narrow Plane).

3.2 Experiment Results

I took my planes, a big ruler, a pen, paper, and my confidence with me, and with pure fire in my eyes, I walked towards the playground. The children who occupied the given area decided to give up their positions, and I started my dirty job. After completing it, I got these results:

Table 1: Flight distances of the wide and narrow airplanes

Launch	Wide (W)	Narrow (N)
1	429	466
2	500	490
3	636	462
4	700	560
5	440	335
6	600	400
7	700	414
8	501	720
9	500	490
10	495	500
11	504	495
12	540	504
13	513	513

I launched my planes from the same spot and measured the distances relative to this spot.

4 Conclusion

To make it easier to compare the distances, let's find the mean of these data:

Wide airplane:

$$\bar{x}_W \approx 542.92$$

$$s_W \approx 88.97$$

Narrow airplane:

$$\bar{x}_N \approx 488.38$$

$$s_N \approx 90.31$$

So:

$$\bar{x}_W > \bar{x}_N$$

In conclusion, according to my calculations and experiment, planes with bigger b , AR , or wingspan have greater F_{net} , which results in a longer flight distance.